

Yi-Chen Zhang, Ph.D.

Research Software Engineer – Autonomous Perception

517-775-9919 chris7462@gmail.com github.com/Chris7462 LinkedIn

Professional Summary

Hands-on Research Software Engineer with 8+ years of experience in autonomous driving perception, combining deep learning research, statistical modeling, and high-performance C++ implementation.

Ph.D.-trained in Statistics with strong foundation in probabilistic modeling, Bayesian inference, and matrix Lie group methods for state estimation. Experienced in developing and evaluating new perception algorithms (object detection, tracking, SLAM) and translating research prototypes into efficient, real-time automotive systems using PyTorch, CUDA, and TensorRT.

Bridges research innovation and production-grade software engineering.

Core Competencies

Perception Research

- 2D/3D Object Detection (FCOS, LiDAR detection, MVXNet)
- Multi-object Tracking (SORT, 3D LiDAR MOT)
- Visual SLAM, Fast-LOAM, OctoMap
- Investigation of transformer-based perception architectures
- Model evaluation, ablation studies, and performance benchmarking

Statistical Modeling & State Estimation

- Bayesian inference and uncertainty modeling
- Error-State Kalman Filter on Matrix Lie Groups
- GNSS/IMU fusion and probabilistic localization
- Linear mixed models and functional data analysis

Software Engineering & Deployment

- C++ (advanced), Python (advanced)
- PyTorch model development and ONNX export
- TensorRT engine build and CUDA-accelerated inference
- Latency profiling and backbone/post-processing optimization
- ROS 2 package development and system integration
- AUTOSAR/MISRA compliance and static analysis (Coverity)

Professional Experience

Isuzu Technical Center of America (ITCA)

2021–Present

Lead Engineer – Autonomous & Artificial Intelligence (2024–Present)

- Conduct research on perception algorithms for autonomous driving, including object detection, tracking, and localization.
- Investigate transformer-based architectures and manifold-based state estimation methods for next-generation perception systems.

- Develop C++ inference backends using NVIDIA TensorRT, reducing FCOS detection latency from 591 ms (PyTorch CUDA) to 34 ms (TensorRT), achieving real-time performance (29 FPS).
- Design modular evaluation pipelines for systematic benchmarking and latency regression testing.
- Build and maintain SLURM-based GPU compute environment supporting large-scale perception experiments.

Senior Engineer – Autonomous & Artificial Intelligence (2022–2024)

- Developed 3D LiDAR object detection and multi-object tracking algorithms for autonomous vehicle platform.
- Implemented GNSS/IMU fusion, wheel odometry, and Visual SLAM for vehicle localization.
- Integrated perception modules within ROS 2-based autonomous stack and conducted on-vehicle validation.
- Mentored junior engineers and supervised research-oriented development projects.

Autonomous Driving Engineer (2021)

- Developed sensor fusion components (camera + LiDAR) for centralized perception system.
- Implemented curb detection and 3D map construction using Fast-LOAM and OctoMap.
- Conducted simulation and real-vehicle integration testing.

APTIV

2018–2021

Algorithm Engineer – Scene Perception & Fused Road Model

- Designed and implemented road geometry estimation and lane centerline prediction algorithms.
- Developed FRM state machine and robust error-handling mechanisms across perception inputs.
- Ensured AUTOSAR/MISRA compliance and implemented unit tests (Google Test, VectorCAST).
- Built analysis pipelines and dashboards for perception performance evaluation.

Selected Research & Technical Contributions

- Developed Error-State Kalman Filter on Matrix Lie Groups for improved vehicle pose consistency (preprint).
- Reduced semantic segmentation inference latency from 406 ms to 26 ms (15× speedup) using TensorRT backend.
- Designed benchmark suites for quantitative evaluation of inference performance (avg/min/max latency).
- Published research in statistical modeling and transportation systems journals.

Publications

- Zhang, Y.-C. (2025+) Error State Kalman Filter on Matrix Lie Group (preprint).
- Zhang et al. (2022) Exploration and Sweeping for Autonomous Sweeper Truck. *Isuzu Technical Journal*.
- Zhang, Y.-C. (2021) Road Geometry Estimation Using Vehicle Trails. *Journal of Intelligent Transportation Systems*.
- Zhang, Y.-C. and Sakhanenko (2019) Naive Bayes Classifier for Functional Data. *Statistics & Probability Letters*.

Education

Ph.D., Statistics

Michigan State University

Dissertation: Functional Data Analysis with Applications to Traffic Flow

M.S., Statistics

National Central University

B.S., Mathematics

National Central University